

Genetics II

GENETIC CODES ARE USED TO MAKE PROTEINS NECESSARY FOR THE BODY.

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Genetic information is held as a code stored on DNA molecules. This code is translated into the many proteins that do the work in a cell. When errors occur in this process, genetic illnesses are possible as a result.

Double helix

A DNA (deoxyribonucleic acid) molecule is a double helix, a ladder-shaped spiral. The “sides” are chiefly ribose sugars, while the “rungs” are made up of four chemical compounds called nucleotides, or bases. The bases are called thymine (T), adenine (A), cytosine (C), and guanine (G). The sequence of these bases is a code that adds up to the instructions for a particular gene.

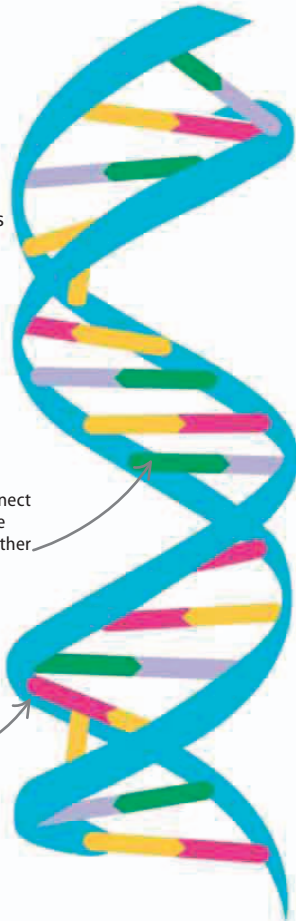
▷ Nucleotide bases

Each rung has a pair of bases. Thymine always pairs with adenine, and cytosine with guanine. Most of the bases are not “read” by the cell, as they do not contain instructions for a gene.

cytosine guanine
adenine thymine

base pairs connect the sides of the helix to each other

the molecule sides are made of sugars



Transcription

The first step in turning a gene into a protein involves making a copy of the gene's DNA stored in the nucleus. This involves transcribing the DNA's code onto ribonucleic acid (RNA). The DNA double helix is unzipped into two unwound strands, and an RNA strand forms next to one of them. The RNA has bases too, but instead of thymine it has a base called uracil (U). The RNA copies the DNA strand and then travels to a ribosome.

▷ Matching up

The bases in the RNA are ordered to match their partners in the DNA, so a cytosine in the DNA is matched by a guanine in the RNA, and uracil matches with adenine.

uracil

noncoding strand

nuclear membrane

coding strand

RNA takes the place of the noncoding strand, and matches the coding strand

